UNDERWATER WONDER

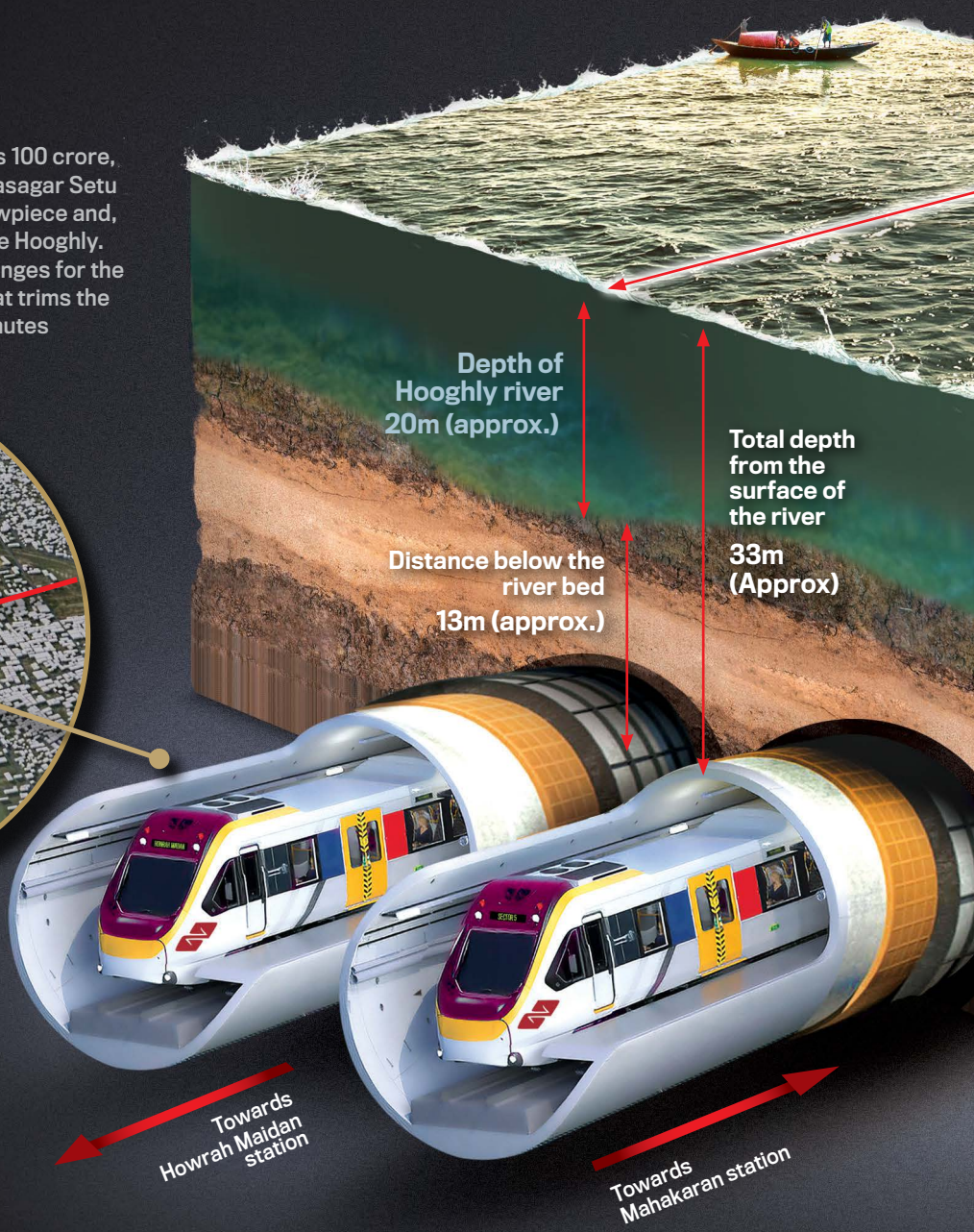
India's first underwater metro rail tunnel beneath the Hooghly river is the highlight of the long-awaited East-West corridor between Howrah and Kolkata. A deep dive...

Graphic by NILANJAN DAS

VITAL LIFELINE FOR KOLKATA

The tunnel, built at a cost of around Rs 100 crore, joins the two iconic bridges–the Vidyasagar Setu and Howrah Bridge–as a Kolkata showpiece and, more practically, as a way to cross the Hooghly. But it presented only one of the challenges for the 16.5-km East-West Metro corridor that trims the Howrah-Salt Lake commute by 50 minutes





MINDING HERITAGE

Before it hit the 520-metre river section, a good part of the 10.8 km underground stretch had to traverse a heritage-rich map around Esplanade. The St Andrews Church (built: 1818), the Currency Building (1841) and two 19th c synagogues–Beth El and Magen David–were to get trains carting 2,000 passengers rumbling past within 100 metres of their foundations. For permissions, the tracks had to be designed with sharp curves, using the latest technology

LEAK PROOF

The inner walls of the tunnel under the river are fortified by six M50-grade, 275 mm thick reinforced concrete segments. Special safety measures were taken to prevent leaks and stop river water from entering the structure. Highly secure concrete mixes were used for the reinforced segments